

LEUKOCYTES

Leukocytes

■ Findings reflect:

- susceptibility of the host
- virulence of organism
- nature and severity of disease process
- systemic response of the individual
- duration of disease process

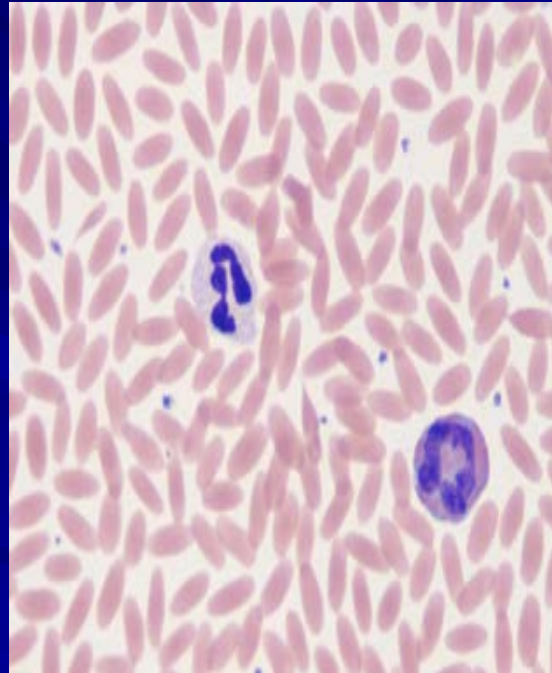
Leukocytes

- Assessment of leukocyte numbers: Total WBC count (all cell types), relative (%) and absolute (cells/uL) differential leukocyte count (WBC separated by type)
- WBC morphologic features: These can give clues as to underlying disease pathogenesis or can identify the cause of the anemia, including parasites

NORMAL MORPHOLOGY OF LEUKOCYTES

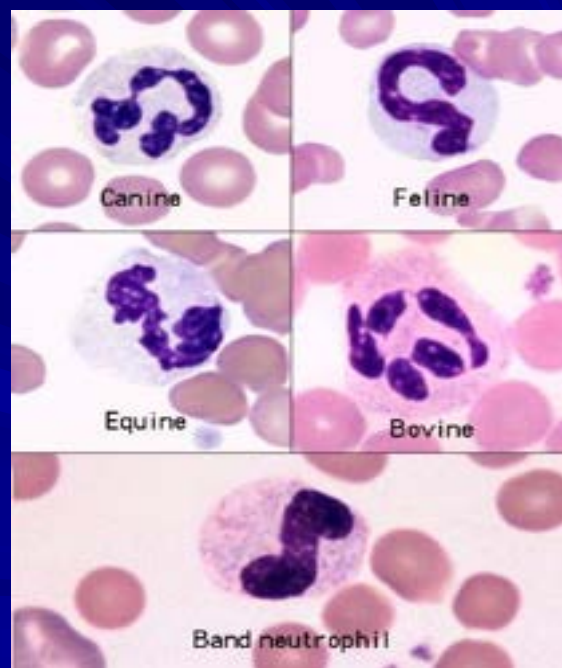
Mature Neutrophils

- **segmented** neutrophil, or the heterophil in birds, reptiles and some mammals, is the predominant granulocyte
- In many species it is the **predominant leukocyte** in health.
- nuclei that are segmented into lobes of condensed chromatin connected by filaments, **mature neutrophils are called segmented neutrophils**
- . Neutrophils are part of the innate immune system and are the **first line** of (and are essential for) defense against **bacterial** pathogens.



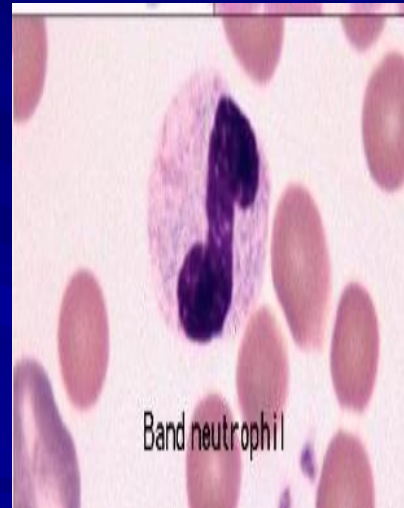
Mature Neutrophils

- Canine neutrophils have **white cytoplasm** that contains **small pink specific** or **secondary granules**.
- Feline neutrophils have cytoplasm that is **white** and **lacks visible granules** ; may contain **1-3 small rod or round blue granules** in their cytoplasm. These are **Döhle bodies** and are actually whorls of rough endoplasmic reticulum on electron microscopy
- Equine neutrophils have **white or slightly pink cytoplasm** with **no visible granules**. The nuclei of equine neutrophils typically are **long, thin and "knobby"** with **clumps of condensed chromatin** projecting from the sides; they may **not have classic segmentation**.
- Ruminant neutrophils have **white cytoplasm** with **small pink granules**; these impart an overall **pink tint** compared to the other species.



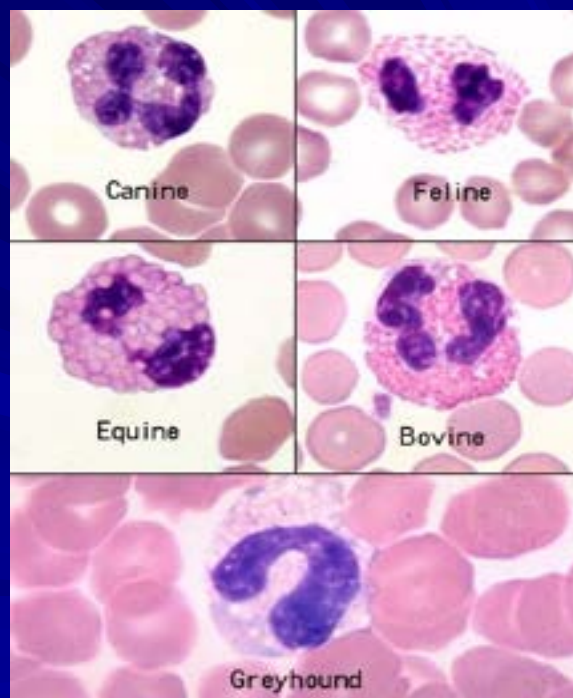
Immature Neutrophils

- immature neutrophils may be seen in peripheral blood and their presence is called a **left shift**.
- indicates a response to **inflammatory cytokines and an inflammatory leukogram**.
- Band neutrophils are distinguished from mature neutrophils by the shape of their nucleus, which **lacks the clear segmentation**



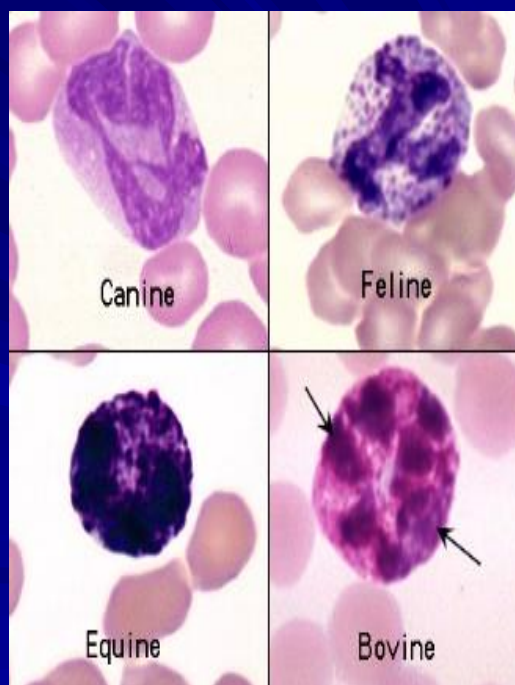
Eosinophils

- dog have **round granules** and a **light blue cytoplasm**.
- cat have **small rod-shaped orange granules** that fill the cytoplasm
- horse have **very large globular orange granules**.
- Ruminant and camelid eosinophils have many **small very round orange granules**. Camelid granules resemble those of ruminants
- A "vacuolated" eosinophil from a Greyhound is also shown. This **lacks visible granules** and mimics a **monocyte** or a **toxic band neutrophil**, except that the cytoplasm contains far more vacuoles than is typical for either of the latter cells; seen in other **sighthounds** and **Golden Retrievers**



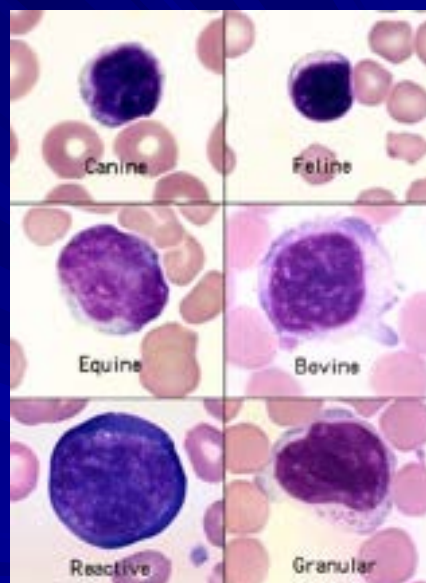
Basophils

- Canine basophils have **pale lavender granules** and a **long and folded ribbon-like nucleus**
- Feline basophils are packed with similar **small, slightly oval pale lavender granules**
- In contrast, equine, bovine and camelid (not shown) basophils have **bright purple granules** (arrows), **like mast cell granules**



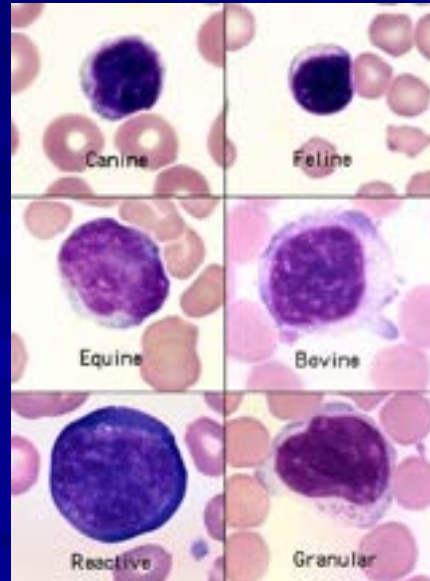
Lymphocytes

- The lymphocytes in the dog and cat are classic small cells with **smudged or smooth chromatin** (heterochromatin) and scant amounts of **light blue cytoplasm**.
- Nuclei are **round to slightly indented or irregular** (as seen in the dog)
- horse and cow are more intermediate in size with **lighter chromatin** but still **clumped** chromatin (heterochromatin) and more **irregular nuclear outlines** (in the horse, at least).



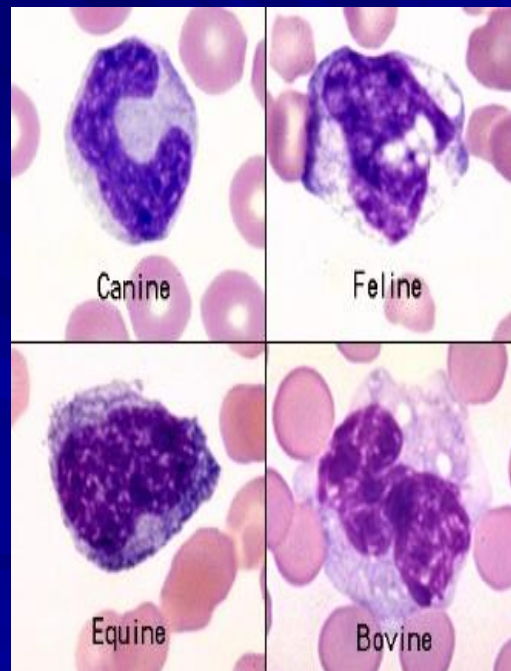
Lymphocytes

- bovine has increased amounts of **light blue cytoplasm** (a variant of normal, not a reactive cell).
- reactive lymphocyte is large with **smooth deep blue cytoplasm** (lots of RNA) and a **single small clear cytoplasmic vacuole**; **chromatin is quite clumped**.
- granular lymphocyte is also from bovine blood, but could be seen in any species; contains a **scattering of red cytoplasmic granules**, located in an indentation of the nucleus; small to intermediate in size and has slightly increased amounts of cytoplasm



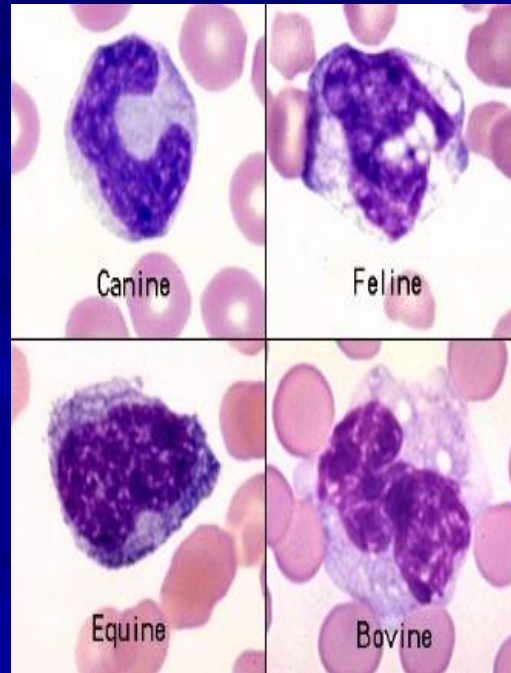
Monocytes

- most problematic for identification, fairly variable in size and appearance.
- often larger than neutrophils and are usually the **largest leukocyte**, but there are no consistent species differences
- nucleus can be **round to kidney-shaped to pseudo-lobulated** (can mimic a neutrophil)
- occasionally be **band-shaped**, especially in dogs, and may be **confused with band neutrophils**



Monocytes

- However, the chromatin of monocytes is **less dense** than neutrophils, and is described as **lacey to slightly clumped**
- cytoplasm is usually smooth and **blue-grey** and may contain a few variably-sized vacuoles, and occasionally a **few very fine pink-red granules**
- Some, like the equine monocyte, resemble lymphocytes; texture of cytoplasm in a monocyte is slightly **grainy or coarse** compared to lymphocyte cytoplasm, which is very smooth and glassy



Normal Leukocyte values

- Predominant blood leukocyte during health in domestic animals
 - cats, dogs, neonatal ruminants : **neutrophils** are the most abundant followed by lymphocytes
 - In mature ruminants: **lymphocytes** are the predominant with fewer neutrophils
 - Horses and pigs: equal numbers of **neutrophils and lymphocytes**

Normal Leukocyte values

■ Cattle

- Influenced by **age** (higher in calves than adult)
- Physiological leukocytosis in beef than dairy breeds
- Low lymphocytes at birth, gradually increases until the age of **4 months** exceeding the number of neutrophils
- Band neutrophils uncommon in healthy cattle
- Slight increase in TLC during **parturition**, **pregnancy**, **stress** and **estrus** (**increase neutrophils** and **decrease eosinophils** on the day of the heat and day after

Normal Leukocyte values

■ horses

- Variation occur in **breeds**; thoroughbred have higher TLC than coldblooded

■ Dogs

- TLC are high in young and decrease with **age**.
- Most of the **decrease is in lymphocytes**; neutrophils remain constant

Normal Leukocyte values

■ Cats

- TLC is difficult to establish due to **excitation** during sample collection
- Physiologic leukocytosis occur under 1 year of age due to emotional stress.
- Neutrophils are in high normal absolute range and **lymphocytes excess** the maximum normal value exceeding neutrophils in number

Normal Leukocyte values

■ Pigs

- **Highest TLC** among domestic animals

■ Goat

- **Age** influence TLC; **lymphocytes was greatest at 6 months of age**

■ Ovine

- similar to cattle

Interpretation of leukocyte counts (physiologic factors)

- Age of the animal
 - Influence both TLC and differential count
 - TLC is high- **dog/calf** at birth; low – **pigs**
 - **Lymphocyte count is > young dog** than adult
 - **Young pig/calf < lymphocytes** than adult
- Breed or species of the animal
 - Predominance of lymphocyte in bovine blood; segmented neutrophils in canine blood

Interpretation of leukocyte counts (physiologic factors)

- Degree of excitement/muscular activity exhibited at the time blood was drawn
 - **Excitement/muscular activity** during collection is not a true state of leukocytes in a patient
 - Marked increased TLC
- Stage of pregnancy
 - (for dogs and cow), increase TLC in the later stages of pregnancy

Interpretation of leukocyte counts (physiologic factors)

- Stage of estrus
 - (cow) slight increase in TLC and neutrophils count on the day of the heat and first day thereafter
- Stage of digestion
 - Influences the TLC, neutrophil counts in **pigs**, **dogs**, minimal effects in **horse**, no influence in ruminants
 - Increase TLC, neutrophils about an hour after eating, **dramatic change in pigs**

Physiologic leukocytosis

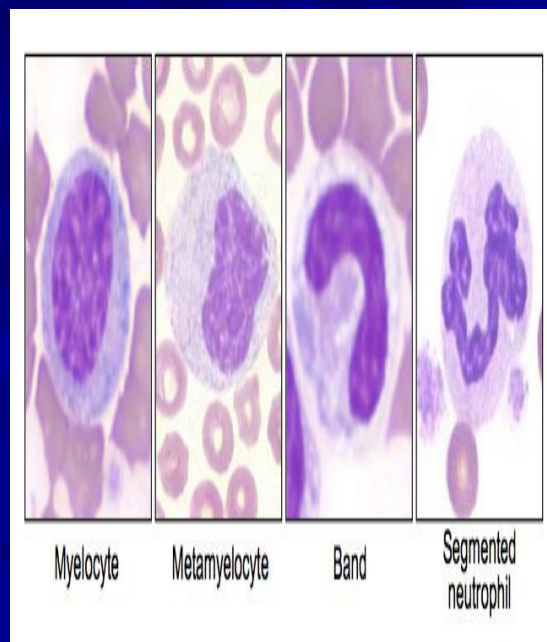
Causes	Cell type
Increased epinephrine w/ fear and excitement	neutrophils, lymphocytes
Exercise	neutrophils, if prolonged, lymphocytes
Anemia	neutrophils
Estrus (cow)	neutrophils, lymphocytes
Digestion (dog, pig)	neutrophils
Stage of pregnancy (dog, cow)	neutrophils

leukogram

leukocytes	stress	physiologic	inflammatory
Total wbc	↑	↑	↓ to ↑
Segmented neutrophils	↑	↑	↓ or ↑
Band neutrophils	absent	absent	Absent or ↑
lymphocytes	↓	↑	↓ to ↑
monocytes	↑	unchanged	↓ or ↑
eosinophils	↓ or ↑	unchanged	↓ or ↑

Shift to the left

- term use do denote an **increase in number of immature neutrophils** in peripheral circulation; indicates an inflammatory leukogram
- due to release of bone marrow stores particularly occurs if the bone marrow reserve of mature neutrophils is low or depleted
- ruminants have lower bone marrow reserve of mature neutrophils than other species and release of the marrow reserve usually occurs in response to acute inflammation

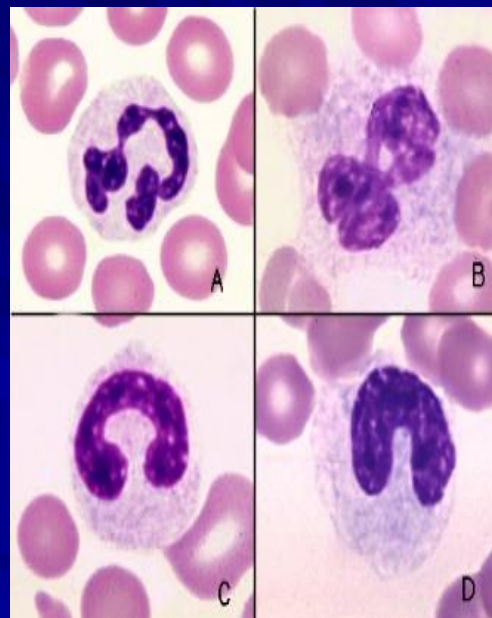


Shift to the left

- Regenerative left shift – **slight to moderate increase in band neutrophils**; convey the concept that they believe there is a **myeloid hyperplasia**, i.e. the bone marrow has had time to and is responding to an inflammatory stimulus
- Degenerative left shift – **normal, low or falling TLC** accompanied by moderate to marked shift to the left. **Immature exceeds mature neutrophils**; generally indicates severe inflammation, which is usually due to **bacterial infection**

Toxic change

- Normally matured segmented and late band neutrophils, shown in panels A and C, have white cytoplasm with pink granules, long and fairly narrow nuclei and tightly condensed chromatin
- Segmented and band neutrophils with toxic change (examples shown in panels B and D) have **less condensed chromatin** than their normal counterparts and **bluer cytoplasm due to retention of ribosomal RNA**

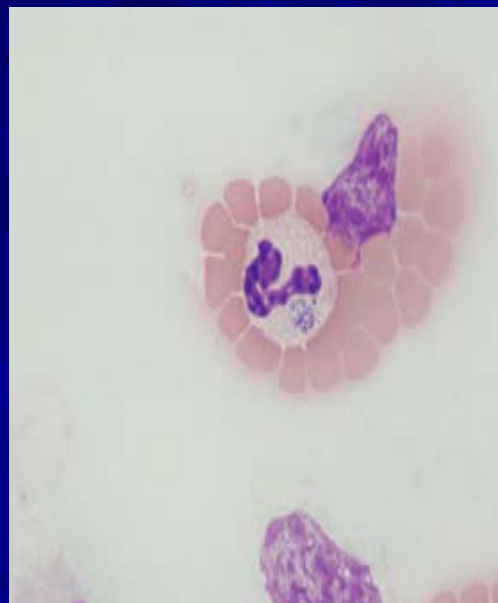


Toxic change (neutrophils)

- (1) **Cytoplasmic basophilia**. The cytoplasm of toxic neutrophils has a diffuse blue color from retained ribosomes. This is the last form of toxic change to resolve with recovery from disease.
- (2) **Cytoplasmic vacuolation**. This change often occurs concurrently with cytoplasmic basophilia, but cytoplasmic vacuolation (foamy appearance) is a more severe manifestation of toxic change. Vacuolation occurs during bacteremia and generalized infection in most species, but is not always specific for infection. This change results from cytoplasmic granule dissolution.
- (3) **Döhle bodies**. These structures are blue to gray, angular, cytoplasmic inclusions representing retained aggregates of rough endoplasmic reticulum. They are most commonly observed in feline blood smears.
- (4) **Toxic granulation**. This form of toxic change is infrequently observed in blood smears of horses and cats, representing severe toxemia. Toxic granulation is characterized by prominent pink-purple granulation of the cytoplasm. Altered membrane permeability of cytoplasmic primary granules permits uptake of Romanowsky stain. Toxic granulation should not be confused with prominent staining of secondary granules, which is not a sign of toxemia.

Infectious agent

- Some infectious agents, including fungi (*Histoplasma*), protozoa (*Toxoplasma*, *Sarcocystis*), bacteria (*Ehrlichia*, *Anaplasma*), can be seen within leukocytes in blood, facilitating diagnosis of these diseases.
- A **grape-like cluster of blue organisms** is seen in this neutrophil at the feathered edge of blood from a horse consistent with *Anaplasma phagocytophilum*.



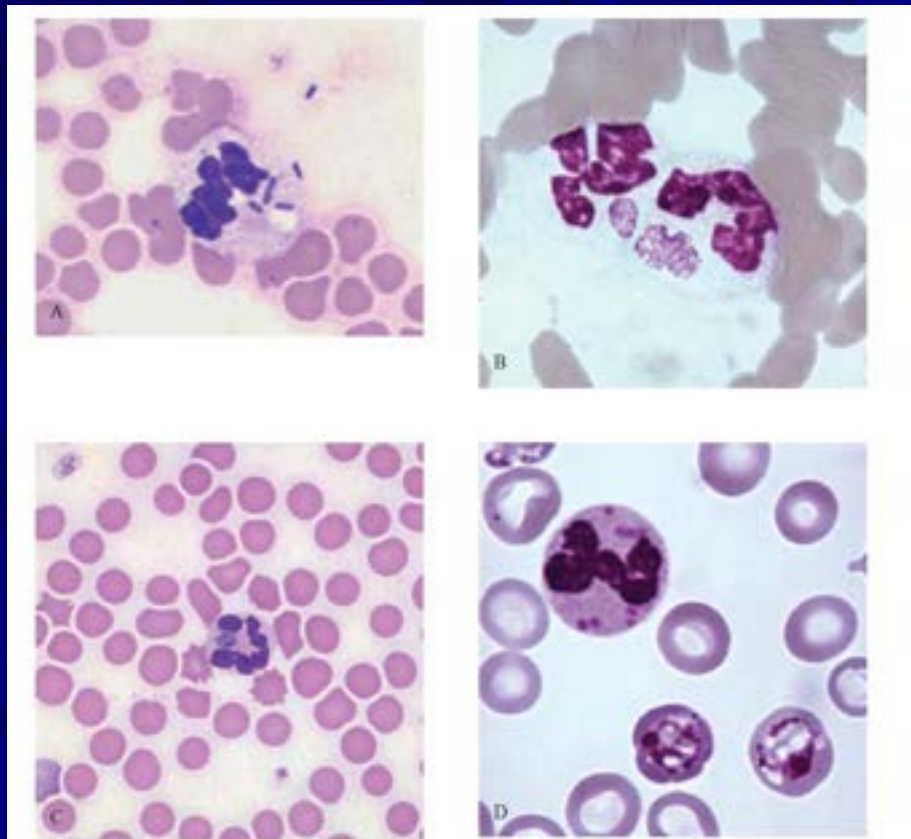


FIGURE 2.3. Infectious organisms within the cytoplasm of leukocytes. A. Equine neutrophil with intracellular and extracellular bacilli (septic foal); B. *Anaplasma phagocytophilum* (previously *Ehrlichia equi*) morulae in cytoplasm of two neutrophils; C. *Ehrlichia ewingii* morula in the cytoplasm of a canine neutrophil; D. magenta canine distemper inclusions in a neutrophil and erythrocytes with traditional Wright-Leishman stain; E. pale blue cytoplasmic distemper inclusion in a canine neutrophil following staining with a rapid modification of Wright stain; F. *Histoplasma capsulatum* yeasts in cytoplasm of two of three canine neutrophils; continued.

Interpretation of leukocyte alteration (severity)

- neutrophilia with slight shift, persistence of eosinophils – mild infection
- High TLC mainly neutrophils – severe condition with good marrow response
- Neutrophilia with lymphopenia and eosinopenia – moderately severe to severe and stress
- Toxic neutrophils – severe
- Immature neutrophils in excess the mature neutrophils – severe
- Falling TLC with diminishing proportion of neutrophils and return of lymphocytes and eosinophils - recovery

Interpretation of leukocyte alteration (duration)

- Acute – either regenerative shift to the left with the characteristic appearance of immature neutrophils/leukopenia
- Chronic – absolute increase in monocytes or degenerative left shift /bone marrow depression
- Leukopenia – overwhelming bacterial infection and endotoxemias; neutrophilia appear if the animal survives

Interpretation of leukocyte alteration (unfavorable prognosis)

- Degenerative shift to the left
- Persistent lymphopenia
- Severe intoxication – toxic neutrophils
- Absence of eosinophils over long period of time
- Extremely high TLC with high % of neutrophils
- Persistent leukopenia with decreased number of all types of cells

Interpretation of leukocyte alteration (favorable prognosis)

- Falling TLC with reappearance of lymphocytes and eosinophils
- Disappearance of toxic neutrophils
- Decrease of immature neutrophils
- Reappearance of eosinophils
- Temporary increase of monocytes
- Regenerative left shift in an animal that has had leukopenia

Leukocytosis

- Physiologic response
- Generalized infection
- Rapidly growing neoplasm
- Acute hemorrhage particularly from body cavities
- Sudden hemolysis of erythrocytes
- Leukemias
- Traums
- Intoxications (drugs, chemicals,venoms)

Leukopenia

- Viral infection such as feline panleukopenia, CD, ICH, feline infectious peritonitis, hog cholera, swine influenza, blue tongue, malignant catarrhal fever, BVD
- overwhelming bacterial infection(gram -)
- Cachexia and debilitating diseases
- X-ray and radiation
- Chemical – chloramphenicol, penicillin, Lead, thallium, mercury, arsenic
- streptomycin, oxytetracycline
- Rickettsial – erlichiosis in dogs and horse
- Shock – endotoxic, septic, anaphylactic

neutropenia

- Excessive emigration to tissues during inflammation (acute bacterial infection, sepsis, necrosis)
- Endotoxin –induced margination (equine salmonellosis)
- Decreased marrow granulopoiesis – feline leukemia virus, feline immunodeficiency virus, parvoviruses
- Cellular infiltration of marrow
- Toxins – bracken fern in cattle, estrogen and sulfadiazine
- Increased destruction of neutrophils –immune mediated neutropenia

neutrophilia

- Glucocorticoids (administration , hyperadrenocorticism)
- Epinephrine (anxiety, physical exertion)
- Pyometra
- Localized neutrophilic inflammatory disease (prostatitis, pancreatitis, peritonitis)
- Hepatozoon infection
- Immune mediated hemolytic anemia

eosinophilia

- Allergic reactions –skin (eosinophilic granuloma complex); respiratory tract (asthma, eosinophilic bronchopneumopathy); intestinal tract (eosinophilic enteritis)
- Parasitism (ecto and endo) in skin, respiratory tract and intestine
- Mast cell tumor
- hypoadrenocorticism

eosinophilia

- Recovery stage of acute infection; re appearance of eosinophils
- Neoplasms of ovary and brain
- Eosinophilic myositis
- Eosinophilic gastroenteritis
- Eosinophilic granuloma in cats
- Association with estrus in some dogs

eosinopenia

- Difficult to detect
- Any stress condition (decrease or complete disappearance
- ACTH administration for therapeutic measure
- Hyperactivity of adrenal gland due to hyperplasia or neoplasia

lymphopenia

- stress from the action of glucocorticoids
- Viral infection affecting bone marrow and lymph nodes
- Chylous effusions caused by CHF or thoracic duct leakage
- Ionizing radiation /immunosuppressive drugs
- Loss of efferent lymph as with rupture of thoracic duct and loss of lymph eg. Enteric diseases
- Intestinal neoplasm
- ACTH administration

lymphocytosis

- Physiologic leukocytosis in cats
- Lymphocytic leukemia
- During recovery stages
- ACTH insufficiency
- Following vaccinations
- Hyperthyroidism

Monocytosis

- chronic diseases
- Monocytic leukemias
- ACTH and corticoid treatment in dog, cow and cat
- associated with stress in dogs and cats
- hyperadrenocorticism